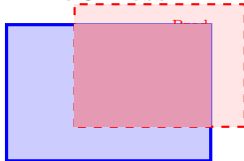


Evolution of Localization Loss Functions

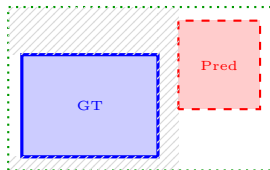
IoU Loss



$$\mathcal{L} = 1 - \frac{|B_p \cap B_g|}{|B_p \cup B_g|}$$

Issue: Zero gradient
when boxes don't overlap

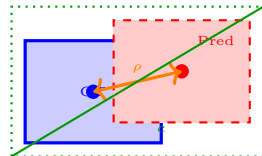
GIoU Loss



$$\mathcal{L} = 1 - IoU + \frac{|C \setminus (B_p \cup B_g)|}{|C|}$$

Adds: Enclosing box
penalty

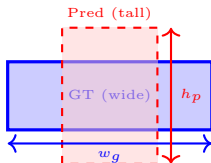
DIoU Loss



$$\mathcal{L} = 1 - IoU + \frac{\rho^2}{c^2}$$

Adds: Center distance
normalized by diagonal

CIoU Loss

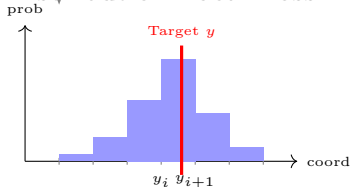


$$\mathcal{L} = 1 - IoU + \frac{\rho^2}{c^2} + \alpha v$$

where v measures aspect ratio

Adds: Aspect ratio
consistency term

Distribution Focal Loss



$$\mathcal{L}_{DFL} = -[(y_{i+1} - y) \log p_i + (y - y_i) \log p_{i+1}]$$

Models localization
as distribution over bins